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Structural and luminescence properties of Sm^{3+} -doped Ca_2SiO_4 phosphors from agricultural waste

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Abstract

Trivalent samarium (Sm^{3+}) ions-doped calcium silicate Ca_2SiO_4 phosphor powders were successfully prepared by conventional solid-state reaction method using SiO_2 and CaO which were obtained from rice husk and egg shells, respectively. The structure of these phosphor powders were studied by X-ray diffraction. Luminescence properties of $\text{Sm}^{3+}:\text{Ca}_2\text{SiO}_4$ phosphors were derived using the excitation, emission and decay curve analyses. The luminescence spectra of these phosphors exhibit intense orange-reddish luminescence at 600 nm due to the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition. The luminescent decay curves for $^4\text{G}_{5/2}$ level of Sm^{3+} ions measured under 404 nm excitation exhibit non-exponential behaviors at higher Sm^{3+} concentrations. The non-exponentiality of the decay curves with shortening of lifetimes were explored by Inokuti-Hirayama model and found that the interaction between Sm^{3+} ions is of dipole-dipole type.

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1. Introduction

Nowadays, the conversion of agricultural waste into low cost luminescent materials is a challenging task to reduce environmental pollution by converting agricultural waste into useful low cost optical materials. In this direction, the present work focuses on synthesis of phosphors by utilizing agricultural waste materials such as rice husk and egg shells [1]. The solid state lighting through phosphor based light emitting devices is deemed to be coming generation light sources because of its low power consumption, higher brightness, fast response, longer durability, environment friendly, etc. [2,3]. Hence, there is a quest to design and develop low cost phosphors from agricultural waste with higher luminescence intensity, good chemical stability and strong absorption in the near-UV region. Selection of host material with low phonon energy and optimized dopant concentration are challenging task because the luminescent efficiency depends on the physical and chemical properties of the host as well as dopant concentration.

Trivalent samarium (Sm^{3+}) ion is paying attention, among rare-earth ions, for higher efficiency orange-reddish luminescence since its luminescent $^4\text{G}_{5/2}$ level possesses higher lifetime of the order of milliseconds, responsible for higher efficiency luminescence in visible and near infrared regions. Hence, Sm^{3+} ion is an excellent activator for developing orange-reddish phosphors. Ju et al. [4] reported Sm^{3+} -doped Ca_2SnO_4 phosphors, which exhibits long lasting orange-reddish phosphorescence for more than 7 h. Haritha et al. [5] studied structural, optical and multipolar interactions in nanocrystalline Sm^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ garnets for light emitting devices. Dalal et al. [6] observed the enhancement in orange-reddish luminescence of $\text{Sm}^{3+}:\text{SrZnV}_2\text{O}_7$ by incorporating charge compensator R^+ ($\text{R} = \text{Li}, \text{Na}, \text{and K}$). Recently, the luminescence properties of Sm^{3+} ion in K_4SrSiO_9 [7], $\text{K}_3\text{Y}[\text{VO}_4]_2$ [8], CaMgPO_4Cl [9], $\text{La}_{0.67}\text{Mg}_{0.5}\text{W}_{0.5}\text{O}_3$ [10] and $\text{Ba}_3\text{Sc}(\text{PO}_4)_3$ [11] have been reported. All these phosphors are synthesized with commercial chemicals. Hence, it is interesting to synthesize $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors from the agricultural waste materials of rice husk ash and egg shells and investigated their structural and luminescent properties for the development of low cost LEDs.

2. Experimental details

$\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ (0.05 and 0.5 mol %) powders were prepared using agricultural waste through solid-state reaction technique. The stoichiometric quantities of CaO from egg shells, SiO_2 from rice husk and Sm_2O_3 (99.99%) from Sigma-Aldrich were weighed and grinded in an agate mortar. Then the mixed powders were sintered at 1250°C for 8 h with several intermediate grindings. Finally these phosphors were calcined at 1250°C for 8 h. The structural properties of these phosphors were derived through X-ray diffractometer (RIGAKU, Miniflex-600) using $\text{Cu } k_\alpha$ radiation ($1.5406 \text{ \AA}$). The photoluminescence excitation and emission spectra, and decay curves were measured using Edinburgh UV-VIS-NIR (FLS 980) spectrometer equipped with xenon arc lamp (450 W) as an excitation source.

3. Results and discussion

The XRD patterns have been measured for undoped and Sm^{3+} -doped Ca_2SiO_4 phosphors and are shown in Fig. 1 for undoped, 0.05 and 0.5 mol % Sm^{3+} -doped Ca_2SiO_4 phosphors. These patterns matches well with the standard ICDD card No. 01-083-0464. This clearly indicates that Sm^{3+} -doped Ca_2SiO_4 phosphors are crystallized in monoclinic structure. The unit cell parameters of $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphor are found to be $a = 5.48 \text{ \AA}$; $b = 6.76 \text{ \AA}$; $c = 9.28 \text{ \AA}$; $\beta = 85.45^\circ$ and volume of $344 \text{ \AA}^3$.

The photoluminescence excitation spectra for Sm^{3+} -doped Ca_2SiO_4 phosphors were recorded for the region 320 - 520 nm by monitoring the emission at 600 nm and are shown in Fig. 2 for 0.05 and 0.5 mol % Sm^{3+} ions in Ca_2SiO_4 phosphors. The excitation spectrum exhibited characteristic absorption bands at about 345, 361, 376, 404, 416, 440, and 477 nm, which are attributed to $^6\text{H}_{5/2} \rightarrow ^4\text{I}_{9/2}$, $^4\text{D}_{3/2}$, $^6\text{P}_{7/2}$, $^4\text{L}_{13/2}$, $^6\text{P}_{5/2}$, $^4\text{M}_{17/2}$ and $^4\text{I}_{13/2} + ^4\text{M}_{15/2} + ^4\text{I}_{11/2}$ transitions of Sm^{3+} ions. It is observed from Fig. 2 that the $^6\text{H}_{5/2} \rightarrow ^4\text{L}_{13/2}$ transition is more intense than other transitions in the excitation spectra. This clearly indicates that $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ (0.05 and 0.5 mol %) could be effectively excited with low cost UV (405) LEDs. Moreover, the intensity of these transitions is increased with increase of Sm^{3+} ion concentration.

The emission spectra for Sm^{3+} -doped Ca_2SiO_4 phosphors measured under 404 nm excitation are shown in Fig. 3. As can be seen from the Fig.3, the spectra consist of emissions at 565, 606, 652 and 704nm, which are ascribed to $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$, $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$, $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{9/2}$ and $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{11/2}$ transitions, respectively. It is clearly noticed that the orange-reddish emission associated with $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition exhibits higher intensity compared to other transitions. As can be seen, the luminescence intensity increases with the increase of Sm^{3+} ions concentration from 0.05 to 0.5 mol %. It clearly indicates that Sm^{3+} ($r = 242$ pm) ions are substituting Ca^{2+} ($r = 240$ pm) ions in Ca_2SiO_4 structure [10,11]. All these results indicate that Sm^{3+} -doped Ca_2SiO_4 could be used as orange-reddish phosphors.

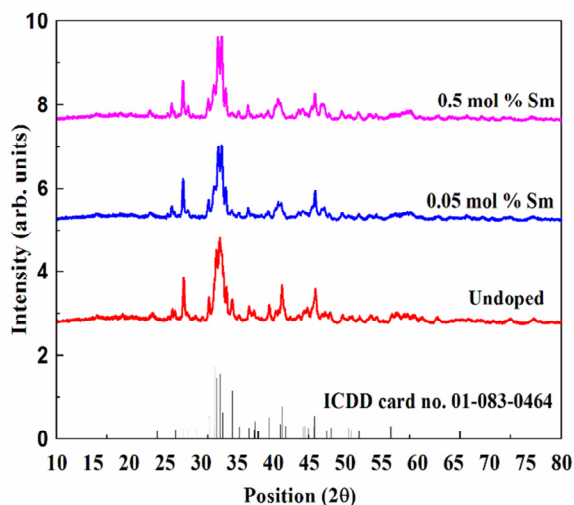


Fig. 1. Powder XRD patterns of $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors.

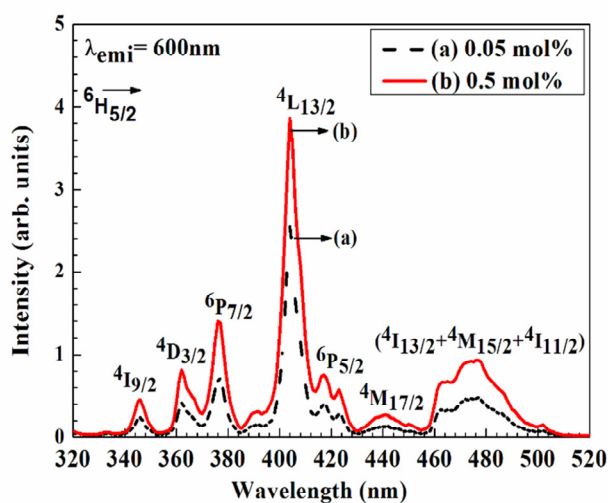


Fig. 2. Excitation spectra for $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors.

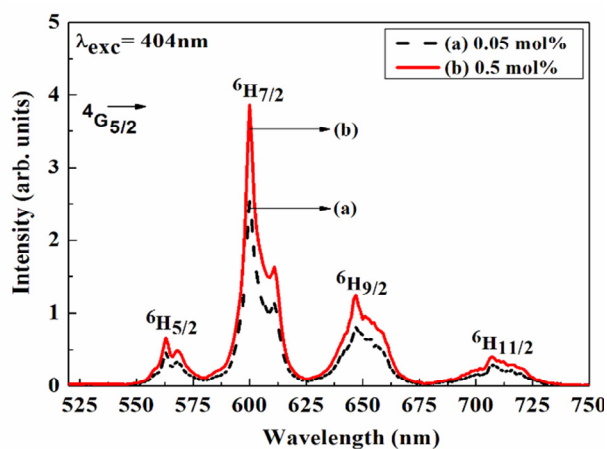


Fig. 3. Luminescence spectra of $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors.

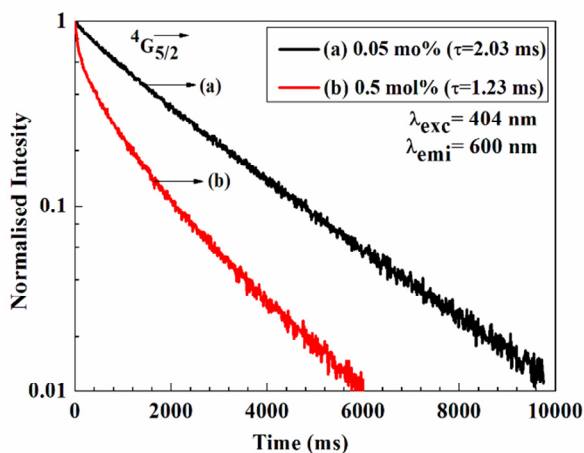


Fig. 4. Decay curves for $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors.

The concentration dependent luminescent decay curves for the $^4\text{G}_{5/2}$ level of Sm^{3+} ions in Ca_2SiO_4 phosphors measured under 404 nm excitation by monitoring the 600 nm emission are presented in Fig. 4. As can be seen, the fluorescence decay curves are exhibiting near single exponential at lower Sm^{3+} concentrations (0.05 mol %). As optically active Sm^{3+} ion concentration increases to 0.5 mol %, the decay curve turns into non-exponential. Hence, the effective lifetime (τ_{eff}) has been evaluated using the following expression.

$$\tau_{eff} = \frac{\int tI(t)dt}{\int I(t)dt} \quad (1)$$

The τ_{eff} for ${}^4G_{5/2}$ level of Sm^{3+} ions in Ca_2SiO_4 phosphors are shown in Table 1. As can be seen from Table 1, the effective lifetimes are found to decrease with increase in Sm^{3+} concentration due to energy transfer between Sm^{3+} ions through cross-relaxation processes [5] and/or energy transfer to luminescence quenching centres (impurities). Similar decreasing trend with increase in Sm^{3+} concentration was reported for Sm^{3+} :phosphors such as $LuGG:Sm^{3+}$ (1.0) [5], $SrZnV_2O_7:Sm^{3+}$ (0.5) [6], $K_4SrSi_3O_9:Sm^{3+}$ (0.5) [7], $K_3Y[VO_4]_2:Sm^{3+}$ (0.6) [8], $La_{0.67}Mg_{0.5}W_{0.5}O_3:Sm^{3+}$ (1.0) [10] and $Ba_3Sc(PO_4)_3:Sm^{3+}$ (0.05) [11]. The non-exponential decay curve of Sm^{3+} -doped phosphors arises due to multipolar interactions between Sm^{3+} ions. To know the type of the multipolar interaction in $Ca_2SiO_4:Sm^{3+}$ phosphors, the non-exponential decay curves have been analyzed with Inokuti-Hirayama (IH) [12] model. According to this model, the temporal fluorescence decay intensity, $I(t)$, is given by

$$I(t) = I_0 \exp \left\{ -\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{3/S} \right\} \quad (2)$$

where 't' is the time after excitation, τ_0 is the intrinsic lifetime. Q is the energy transfer parameter which depends on the value of 'S' (= 6 for dipole-dipole, 8 for dipole-quadrupole, 10 for quadrupole-quadrupole interactions). As can be seen from Fig. 4, the best fit was found to be for $S = 6$ with $Q = 1.57$ for 0.5 mol % of Sm^{3+} ions, which confirms that dipole-dipole interaction are responsible for energy transfer between Sm^{3+} ions with shortening of lifetimes.

The color coordinates are necessary to know the nature of color emitted by phosphors and hence they have been determined using the emission spectra. It is observed that the color coordinates (x, y) fall in orange-red region (see Fig. 5) and remains almost same for 0.05 and 0.5 mol % $Sm^{3+}:Ca_2SiO_4$ phosphors as shown in Table 1.

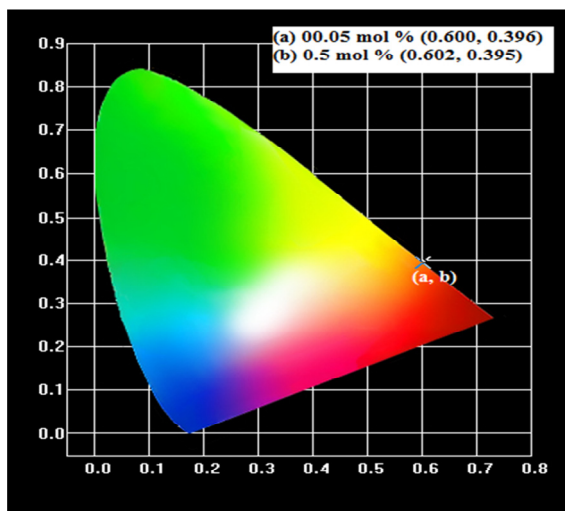


Fig. 5. CIE diagram for $Ca_2SiO_4:Sm^{3+}$ phosphors.

The color co-ordinates and lifetimes for the present and reported systems are compared in Table 1. As can be seen from Table 1, the color co-ordinates and lifetimes are obtained for the present $Ca_2SiO_4:Sm^{3+}$ phosphors synthesized from agricultural waste are comparable to those of the reported systems synthesized from commercial chemicals [5-11]. Hence, the $Ca_2SiO_4:Sm^{3+}$ phosphors synthesized from the utilization of agriculture waste could be useful for the fabrication of low cost LEDs.

Table.1 Color coordinates (x, y) and lifetimes (τ , ms) for Sm^{3+} -doped phosphors.

S. No.	Host (mol %)	(x, y)	τ	Reference
1	$\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ (0.05)	(0.600, 0.396)	2.03	[Present work]
2	$\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ (0.5)	(0.602, 0.395)	1.23	[Present work]
3	$\text{LuGG}:\text{Sm}^{3+}$ (1.0)	-	2.50	[5]
4	$\text{SrZnV}_2\text{O}_7:\text{Sm}^{3+}$ (0.5)	(0.546, 0.449)	2.89	[6]
5	$\text{K}_4\text{SrSi}_3\text{O}_9:\text{Sm}^{3+}$ (0.5)	-	2.46	[7]
6	$\text{K}_3\text{Y}[\text{VO}_4]_2:\text{Sm}^{3+}$ (0.6)	(0.562, 0.435)	1.24	[8]
7	$\text{CaMgPO}_4\text{Cl}:\text{Sm}^{3+}$ (0.5)	(0.62, 0.37)	-	[9]
8	$\text{La}_{0.67}\text{Mg}_{0.5}\text{W}_{0.5}\text{O}_3:\text{Sm}^{3+}$ (1.0)	(0.772, 0.277)	0.75	[10]
9	$\text{Ba}_3\text{Sc}(\text{PO}_4)_3:\text{Sm}^{3+}$ (0.05)	(0.568, 0.424)	1.64	[11]

4. Conclusions

The $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors have been prepared by conventional solid-state reaction method and characterized for their structural and photoluminescence properties. The powder X-ray diffraction patterns indicate that the synthesized Sm^{3+} -doped Ca_2SiO_4 phosphors are crystallized in monoclinic structure. The color co-ordinates obtained from emission spectra are found to fall within the orange-reddish region. The decay curves exhibit single exponential at lower concentration and turns into non-exponential at higher concentrations accompanied by a shortening of lifetime. The non-exponential decay curves are well fitted for $S=6$ and indicates that dipole-dipole interactions are responsible for the energy transfer. The luminescence properties reveal that the $\text{Ca}_2\text{SiO}_4:\text{Sm}^{3+}$ phosphors could be suitable candidate for the development of low cost orange-reddish light emitting devices.

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